CESifo Area Conferences 2017

CESifo Conference Centre, Munich

CESifo Area Conference on

Applied Microeconomics

17–18 March

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The Diffusion of New Institutions: Evidence from the Renaissance Venice's Patent System [1]

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VERY PRELIMINARY DRAFT

March 9, 2017

[1] We are indebted to Roberto Berveglieri for numerous discussions about the Venetian patent systems, and to Giovanni Caniato, Luciano Pezzolo and Andrea Zannini for providing insightful details of the Venetian guild system. We thank Michel Serafinelli for helpful comments. We thank Lavinia Parziale for providing access to guild data.

Abstract

What factors affect the diffusion of new economic institutions? This paper examines this question exploiting the introduction of the first regularized patent system which appeared in the Venetian Republic in 1474. We begin by developing a model which links patenting activity of craft guilds with provisions in their statutes. The model predicts that guild statutes that are more effective at preventing outsider's entry and at mitigating price competition lead to less patenting. We test this prediction on a new dataset which combines detailed information on craft guilds and patents in the Venetian Republic during the Renaissance. We find a negative association between patenting activity and guild statutory norms which strongly restrict entry and price competition. We also show that guilds which originated from medieval religious confraternities were more likely to regulate entry and competition, and that the effect on patenting is robust to instrumenting guild statutes with their quasi-exogenous religious origin. We find that patenting was more widespread among guilds geographically distant from Venice, and among guilds in cities with lower political connection which we measure exploiting a new database on noble families and their marriages with members of the Great Council. Our analysis suggests that local economic and political conditions may have a substantial impact on the diffusion of new economic institutions.

Keywords: patents, competition, guilds, institutions.
JEL classification: O33, O34, K23

1 Introduction

The social, economic, legal and political organization of a society, its "institutions", is a primary determinant of economic growth (Acemoglu and Johnson, 2005). Institutional change requires both the development and the diffusion of new socially beneficial institutions. In particular, the impact of new economic institutions - such as novel forms of contracts or property rights – will depend on the rate at which they are adopted and displace older institutions (Acemoglu et al., 2005). The development literature has identified a variety of factors explaining the international diffusion of new institutions, such as human capital (Glaeser et al., 2004), culture (Tabellini, 2010) and geography (Comin et al., 2012). On top of these national variables, the rate of adoption of novel economic institutions is also likely to be affected by local conditions - such as industry composition or the presence of elites – which vary at a much narrower geographical level such as a region or a city. The empirical evidence on the role of these micro-level factors is scarce.

In this paper, we aim to address this gap by studying the pattern of local diffusion of one of the main economic institutions which governments use to increase innovation incentives and spur economic growth: patent rights. Patents provide temporary monopoly rights on an innovation which prevent expropriation and support private contracting over the new technology. The innovation literature has documented a large variation in the rate of patenting across industries and in the perceived effectiveness of patents across firms (Levin et al., 1987; Cohen, Nelson, and Walsh, 2000). These findings have typically been interpreted as suggesting that the social and economic value provided by intellectual property rights is highly heterogeneous. To understand the roots of this heterogeneity - i.e. why some inventors choose to heavily rely on patents and why others do not- is essential for the design of patent policies. If, for example, a substantial share of innovation occurs in industries in which patents do not play an important role, policies that strengthen intellectual property rights may do little to raise the overall level of innovation (Machlup and Penrose, 1950; Moser, 2012). Similarly, when only few industries rely heavily on patent rights, changes in patent policies may dramatically affect the direction of technical change (Moser, 2005). Finally, if the effects of patent rights are highly heterogeneous across firms and industries, it is likely that a one-size-fits-all patent system as the one currently in place is second best (Acemoglu and Akcigit, 2012).

This paper provides new insights on the determinants of patenting activity, exploiting the introduction of the first regularized patent system which appeared during the Renaissance in the Venetian Republic. In 1474 the Venetian Senate passed a patent act which regulated the granting of patents for novelty, ingenuity, and utility. The predominant view among patent law historians is that this act formalized the pre-1474 practice of "ad hoc" government granted privileges with an administrative-centered system, strikingly similar to the modern Anglo-American system (Merges and Duffy, 2013). The patents awarded in the late fifteenth century in the Venetian Republic provide a unique opportunity to study the diffusion of a new institutional form of property rights. The distinctive nature of the Venetian Patent Act (Mandich, 1936) allows us to examine the introduction of a drastically new innovation policy, rather than focusing on marginal changes of modern policies. Moreover, the historical nature of our data is useful to understand whether heterogeneity in the use of patent rights is a persistent phenomenon, or is a more recent issue linked to modern technology trends.

We begin our analysis with a simple theoretical model which describes the patenting decision of inventors at the time of the Venetian Republic. The theoretical framework highlights two key differences between the modern patent regime and the Venetian system. First, Venetian patent rights provided not only the 'negative' rights to exclude through monopoly power, but also the 'positive' right to enter into craft guilds for innovators which were not guild members (Mandich, 1948; Sichelman and O'Connor, 2012). Second, guilds had the power to oppose and block patent applications (Berveglieri, 1995;1999; Trivellato, 2008). We show that the interplay of these two features implies that the level of patenting can vary substantially across guilds, and that this is true both for guild members and for outside inventors. More specifically, the model shows that the level of patenting in a technology area is strongly related to the ability of guild statutes to prevent entry of outsiders and to mitigate competition among guild members. Greater statutory restrictions allow guild members to extract high rents, and this increases their incentives to prevent patenting by other members and external innovators.

Our empirical analysis exploits a new dataset which combines detailed digitized information on craft guilds in Italian cities for the period 1200-1800, with data on the patents granted by the Senate of the Venetian Republic after the 1474 patent act. The main findings are as follows. First, we find a strong negative association between the patenting in the technology sector of a guild and the presence of detailed internal rules limiting entry and competition. Results are robust to including a variety of controls for city and guild characteristics, and to using alternative methodologies to impute patents to guilds.

Second, we show that patenting was more frequent for guilds located in cities geographically distant from Venice. Moreover, we construct a measure of political connection at the city level exploiting a unique database on Venetian nobility and marriages between noble families and members of the Great Council. We document a negative association between political connections and patenting. We interpret these findings as suggesting that intellectual property rights were a substitute to other forms of formal and informal protection.

Finally, we provide evidence of the causal nature of the negative effect of guilds' internal statute on patenting by exploiting an instrumental variable which relies on the religious origin of some of the guilds in our sample. In fact, a number of the guilds in Northern Italy originated from medieval religious confraternities a couple of centuries before the patent act. We show that the religious origin of the guild is a strong predictor of statutory provisions restricting entry and competition. At the same time, we discuss how the religious origin of the guild is likely to be driven by idiosyncratic reasons and is uncorrelated to other observable guild characteristics. The instrumental variable analysis confirms the negative relation between patenting and the strength of guilds' statutes.

These findings are connected to the economic history literature on the role of craft guilds. A common view is that medieval craft guilds were technophobic (North, 1981). More recent studies provide a more nuanced view, recognizing that there were substantial differences across guilds, and some of them were much more receptive than others to novelties and technological advances (Epstein, 2004). In her analysis of Venetian silk and glass production, Trivellato (2008) emphasizes the crucial role of intra-guild interactions which was often shaped by guild statutes, with experimentation taking place only when statutory norms were not too restrictive. Our empirical analysis confirms Trivellato's thesis, and highlights the fundamental role of intra-guild interactions.

Our paper is also related to studies that investigate the effects of occupational licensing. Kleiner (2000) provides a survey of the literature. Persico (2015) develops a theory showing how internal politics of a licensing association can lead to expansion of the licensure. Our analysis illustrates how occupational licensing and self-regulation may interact with the diffusion of new economic institutions. The role of internal rules and how they influence technology adoption is also the focus of Bridgman (2015) who studies why unions may favor restrictive work regulations and how these regulations may induce resistance to technology adoption. Finally, our paper adds to the literature on the relationship between competition and innovation (Bloom et al, 2005; Cohen, 2010). Our findings suggest that the competitive structure may not only affect the level of innovation but also the propensity to rely on patent protection.

The paper is organized as follows. Section 2 provides a brief description of the origin and functioning of the Venetian patent system. Section 3 develops a model showing the link between guild statutory norms and patenting. Section 4 describes the data and discusses the econometric specification. Section 5 presents the empirical results. Concluding remarks briefly summarize the findings and their implications.

2 Renaissance Venice and its patent system

This section provides a brief historical overview of the Venetian Republic between the fifteenth and sixteenth century, and illustrates the main features of the 1474 patent act.

2.1 The Venetian republic in the XV and XVI century

During the period of our study, the "Serenissima" Republic of Venice was one of the largest regional economy of Renaissance Europe. Its center was the maritime city of Venice with roughly 100,000 inhabitants in 1500, about half of the population of north-east Italy at that time (Malanima, 1998). The Venetian state extended on the "Terraferma" dominion, a compact and densely populated area which included large cities such as Verona and Vicenza. Figure 1 (from Knapton, 2013) illustrates the State boundaries around the period of our study. A number of additional cities in the Greek peninsula and in South-East Europe, such as Corfu, Andros and Cyprus were also under the control of the Venetian Republic and were instrumental ports for long-distance trade between Western Europe and the Levant (Borelli, 1980).

The Venetian Republic was based on a careful balance of power that originated by an attempt to restrain the power of a single person or governing body and led to remarkable political stability (Lane 1973). Membership to the main governing institutions was precluded to lower classes such as artisans and shopkeepers. Moreover, following the Serrata or "closure" in 1297, political functions were restricted to a hereditary nobility that had the exclusive right to sit in the Great Council, the legislative assembly of the Republic. Because of the large size of the Great Council, most of legislative functions were delegated to the Senate, a smaller assembly (about 300 senators) elected by the Great Council (Borelli, 1980). Some members of the Senate had the right of legislative initiative ('metter parte') others were only entitled to vote ('metter ballotta'). Among the senators entitled both to vote and to propose new laws there were three "Provveditori di Comun" who also oversaw transport infrastructures and mercantile trade (Borelli, 1980; Zaggia, 2004; Di Stefano, 2011). The Doge was the personal embodiment of the Republic, it was elected by a committee of 41 nobles chosen by the Great Council. In 1474 the Doge was Nicolo' Marcello and eleven Doges took office between 1474 and 1550 (Rendina, 1984).

At the time of our study, the main threat to Venice's trade supremacy and the preservation of its economic power was the Ottoman Empire which was expanding dramatically under the leadership of sultan Selim I (Borellli, 1980). Moreover, the 1492 discovery of America started shifting the center of long-distance trade away from the Mediterranean toward the Atlantic.

The economy of the capital was driven by the vast trading activity in spices, dying materials, silk, cotton, slaves, and precious metals (Pezzolo, 2013). On top of this vibrant trade, artisan production also flourished both in Venice and in Terraferma. The Arsenal was one of the largest industrial sites in Europe, and glassmaking was among the most prestigious urban occupations at the time (Trivellato, 2008). The mainland was marked by a lively wool and silk production (Demo, 2013).

Merchants and craftsmen were organized in guilds, self-governed organizations which controlled various aspects of the economic activity. Guild statutes prescribed technical characteristics of products, and regulated entry, apprenticeship and competition. Often they precluded guilds members from moving to another city (Belfanti, 2004). The Venetian government fostered guild membership for fiscal reason, and about 20 percent of the population of the city of Venice belong to a guild.[1] Guild members were excluded from government, but the Venetian constitution guaranteed them the right of judicial appeal against the government and guild officers (Lane, 1973).

2.2 The 1474 patent act

On March 19 1474 the Venetian Senate passed an act or "parte" regulating the granting of patents. While there is evidence that a small number of 'ad hoc' privileges for new inventions and mineral extraction were granted by the Venetian government before this act (only five patents according to Mandich, 1936), the 'parte' of 1474 is the very first law regulating the patent application and granting process, and has been recognized by numerous historians and law scholars as the legal foundation of the modern patent system (inter alia see Mandich 1948;

[1] This share remained stable, with minor fluctuations, from the XVI Century till 1797, the end of the Venetian state (Costantini, 1987).

Duffy, 2007; Golden, 2013).

The act indicates that patent applications or 'suppliche' were to be notified to the 'Provveditori di Comun' and the patents (or 'privative') were typically granted after Senatorial approval (Berveglieri, 1995). The subject matter to be patented was required to be a "new and ingenious device" and the effect of a patent was to stop "every other person in any of our territories and towns to make any further device conforming with and similar to said one without the consent and license of the author" (Mandich, 1948). The novelty content was evaluated on the basis of the technical knowledge available in the Venetian dominium, implying that a patent could be granted to products or processes already in use elsewhere (Molà, 2014). The patentee was required to implement the invention ('messa in opera') within a specified period of time. In case the patentee failed to use the device, the patent was revoked (Berveglieri, 1995).[2]

There are three main features of the Venetian patent system which are key to our analysis. First, patents could be granted to all inventors regardless of their citizenship status or guild membership. Thus, patents were both 'negative' rights to exclude but also 'positive' rights to practice the invention and operate in industries controlled by guilds (Mandich, 1948; and Sichelman and O' Connor, 2012). For example, Florentine inventor Cosmo Scatini was granted a patent for high quality black silk dying which permitted him to enroll in the dyer's guild (Belfanti, 2004).

Second, guilds were often involved in the patent granting decision process, through the evaluation of the novelty content. This examination involved, most of the time, a test of the new product or process (the "experienza") to verify, before granting the patent, whether the invention was actually working. Trivellato (2008) describes the opposition of the Venetian silk spinners' guild to the patent application of Iseppo Giovan Perin Mattiazzo for a new hydraulic mill for spinning and throwing silk.[3]

Third, patent holders were expected to share the technology with guild members with the payment of an appropriate licensing fee. Such licensing requirement is often mentioned in the patent records, without specifying the precise amount but requesting a "discrete sum" of money for the transfer or payment of "adequate reward" (Belveglieri, 1995).

[2] The act established a patent length of 10 years, but was common for applicants to request longer protection. Mandich (1936) describes cases in which patent rights lasted 25 and even 70 years.

[3] Mola' (2000) reports a number of other opposition cases, such as the 1583 spinning machine patent of Urbano Bonturelli and the 1597 silk bleaching patent of Giacomo di Bianchi and Innocente Soardo.

While a number of historians have examined the administrative details of the Venetian patent system and collected detailed information on patent records, very few studies have addressed the question of why the Senate passed the patent act in 1474. Lane (1973) and May (2002) suggest that the growing economic and trading power of the Ottoman empire and Antwerp led Venetian policy makers to focus on industrial activities. Berveglieri (1995, 1999) and Belfanti (1995) emphasize the goal of attracting foreign inventors to the Venetian Republic to compensate for the lost supremacy of Venetian guilds in various industrial sectors.[4] Mandich (1936) suggests that successful experimentation with monopolies in mineral rights may have led Venetian authorities to legislate on patent rights.

3 Theoretical model

In this section, we develop a simple theoretical model to study patenting incentives in the Venetian Republic and examine the effects of guild statutes on firms' patenting strategies.

3.1 Set-up

Consider an industry with three firms and two periods $t = 1, 2$. Two firms belong to a guild, and the third firm is an outsider. In the absence of innovation, guild members sell a standard product to consumers. The surplus created by the standard product is π per period. We assume that the guild can appropriate a fraction $\alpha(\theta)$ of this surplus, with $\alpha'(\theta) > 0$. The appropriated surplus is shared equally among guild members. The parameter θ captures the strength of the guild's internal statute, with a larger value of θ indicating larger collusive power among the members which allows greater profit extraction.

At $t = 1$ one of the firms develops an innovation which can increase the surplus to $\pi + \Delta$ per period. Innovations are distributed with cumulative distribution $F(\Delta)$ with support $[0, \infty]$. To patent the innovation costs c and patent protection lasts for one period. The patent grants the innovator the right to extract the full surplus for the period. The patent holder negotiates licensing deals with the other guild members by making take-it-or-leave-it offers to them. Once the patent has expired, the technology becomes freely available to all guild members.

The outsider firm cannot enter the guild without an innovation. Entry is guaranteed if the outsider firm obtains a patent. If it innovates but does not obtain a patent, entry occurs

[4] Belfanti (2004) documents how the view that attracting talented people to a city was crucial for economic growth was wery popular among intellectuals of the time.

with probability $\beta(\theta)$ with $\beta'(\theta) < 0$, which captures the idea that the stronger guild statutes are, the more difficult it is for an outsider to enter.

Each guild member can oppose a patent application by paying an opposition cost, κ. If the patent is opposed, the technology is appropriated and shared among all the guild members. If the patent of the outsider is opposed, its entry is blocked as well.

For simplicity we set $\alpha(\theta) = \theta$ and $\beta(\theta) = 1 - \theta$ (for robustness see section 3.4). We also assume that $\pi > 3c$ to focus on the cases in which the cost of obtaining a patent is not too large relative to total surplus.

3.2 Innovation by a guild member

We start with the case in which a guild member is the innovator. With a patent on the innovation, the guild member profits are equal to

$$\pi + \Delta - \frac{\theta\pi}{2} + \frac{\theta(\pi + \Delta)}{2} - c.$$

Specifically, in the first period patent protection allows the firm to extract the full surplus $\pi + \Delta$. At the same time the licensing negotiation with the other member implies that $\frac{\theta\pi}{2}$ is transferred through licensing.[5] In the second period, the innovation is shared by the two members, each obtaining $\frac{\theta(\pi+\Delta)}{2}$. If the innovator chooses not to patent, its profits are $\theta(\pi + \Delta)$ because the technology is shared for two periods. Patenting is more profitable than not patenting only if

$$\pi + \Delta - \frac{\theta\pi}{2} + \frac{\theta(\pi + \Delta)}{2} - c > \theta(\pi + \Delta)$$

or

$$\Delta > \widetilde{\Delta}(\theta) = \frac{2}{2 - \theta}\left(c - \pi(1 - \theta)\right).$$

Notice that $\widetilde{\Delta}(\theta) > 0$ only if θ is large enough. Moreover $\frac{d\widetilde{\Delta}(\theta)}{d\theta} > 0$. This implies that as the strength of the internal statute increases, guild members patent only their more valuable innovations and the propensity to patent decreases in θ.

Consider now the incentives of one guild member to oppose the patent by the other guild member. Opposition is profitable if

$$\theta(\pi + \Delta) - \kappa > \frac{\theta\pi}{2} + \frac{\theta(\pi + \Delta)}{2}$$

[5]The implicit assumption here is that in case of disagreement the innovation is not implemented for one period until the patent is expired, so that each firm gets $\theta\pi/2$. Results are robust to considering alternative outside options.

which is satisfied if

$$\Delta > \widehat{\Delta}(\theta) = \frac{2}{\theta}\kappa.$$

Notice that $\frac{d\widehat{\Delta}(\theta)}{d\theta} < 0$ which implies that guild members block patents of other guild members more often as the strength of the internal statute increases.

The above discussion implies that the likelihood of patenting goes down as we increase the internal strength of the guild statute, because guild members are less likely to apply for a patent and they are also more likely to block a patent of other members. Conditional on an innovation of a guild member, the likelihood of patenting is

$$P(\theta) = F(\widehat{\Delta}(\theta)) - F(\widetilde{\Delta}(\theta))$$

which decreases in θ.

3.3 Innovation by an external innovator

The profits of an external innovator with a patent are

$$\pi + \Delta - \theta\pi + \frac{\theta(\pi + \Delta)}{3} - c.$$

In the first period the innovator extracts full surplus and strikes licensing deals with the guild members offering $\theta\pi/2$ to each of them. In the second period the innovation is shared among the three firms. Without a patent, the external innovator enters the guild with probability 1-θ and the technology is immediately shared with the guild members. Patenting is more profitable than entering without patent if

$$\pi + \Delta - \theta\pi + \frac{\theta(\pi + \Delta)}{3} - c > (1 - \theta)\frac{2\theta(\pi + \Delta)}{3}$$

that occurs if

$$\Delta > \widetilde{\Delta}_E(\theta) = \frac{3c - 3\pi + 4\pi\theta - 2\pi\theta^2}{3 - \theta + 2\theta^2}.$$

Notice that $\widetilde{\Delta}_E(0) = (3c - 3\pi)/3$ and $\widetilde{\Delta}_E(1) = (3c - \pi)/4$. Moreover, because $\pi > 3c$ we have that $\frac{d\widetilde{\Delta_E}(\theta)}{d\theta} > 0$ and $\widetilde{\Delta}_E(\theta) < 0$ for each θ, which implies that the external innovator always patents, no matter the strength of the guild statutes. For low values of θ patenting is optimal since the innovator appropriates a large share of the profits generated by the innovation during the first period. When θ is large, patenting is optimal to overcome the difficulties of being admitted in the guild.

By opposing the patent, guild members prevent entry and appropriate the rent from the new technology from period one. Without opposition, each guild member receives licensing rent ($\pi\theta/2$) for one period and shares the technology with the other two firms in the second period. Opposing the patent is more profitable than accommodating entry if

$$2\theta\frac{(\pi+\Delta)}{2} - \kappa > \frac{\pi}{2}\theta + \frac{\theta(\pi+\Delta)}{3}$$

or

$$\Delta > \widehat{\Delta_E}(\theta) = \frac{3}{2\theta}\left(\kappa - \frac{\pi}{6}\theta\right)$$

which is decreasing in θ, i.e. opposition is more likely with high θ. Conditional on the outsider innovating, the likelihood of patenting is

$$P(\theta) = F(\widehat{\Delta}_E(\theta))$$

which is also decreasing in θ.

3.4 Discussion

Our simple model illustrates how the propensity to patent in a technology area is affected by the strength of the statutes of the guilds operating in the field. As the strength of the statute increases, the collusive power of a guild goes up, and the value of the monopoly rent generated by the patent decreases. Thus, strong statutes reduce patenting incentives of guild members. Moreover, statute strength allows guild members to extract high rents from the technologies that they appropriate through patent opposition. This implies that in the presence of strong statutes patents by guild and non-guild members are more likely to be opposed. Together, these two effects generate the testable prediction that patenting activity is likely to be less prominent in technology fields in which guilds have strong statutes.

There are two assumptions in the model that warrant additional discussion. First, to obtain a closed form threshold for the patenting and opposition strategies we set the impact that guild statutes have on rent sharing and entry equal to $\alpha(\theta) = \theta$ and $\beta(\theta) = 1 - \theta$. In the appendix, we show that our comparative statics are robust to considering more general functions $\alpha(\theta)$ and $\beta(\theta)$. Specifically, we show that our comparative statics hold under mild assumptions on these functions and derive a sufficient condition that generalizes our main results. Second, our baseline setting assumed that the patentee has full bargaining power during the licensing negotiations and that it can appropriate the whole surplus of the innovation (while the other guild members obtain the status-quo profits $\theta\pi/2$). In the appendix, we relax this assumption and study a more general set-up in which the surplus is shared through Nash bargaining. We show that our main results are robust, as long as the bargaining power of the patentee is not too small.

Finally, a notable feature of the Venetian patent system is the opportunity for guild members to oppose patent applications. This resembles modern administrative processes at the European and United States Patent Offices (Harhoff and Reitzig, 2004; Hall and Haroff, 2004). Our simple model suggests that these opposition systems may have a variety of effects on entry and patenting behavior. Venetian patents granted external innovators not only short-term monopoly rights but also the right to enter in a guild and extract long term rent. This induces external innovators to patent all their inventions, even those with very low value (i.e. $\Delta < c$). Through opposition, incumbent guild members have the ability to screen out such inefficient patenting. At the same time, opposition allows non-innovating guild members to protect their short term and long term rents. This creates an incentive to block entry of valuable technologies by external innovators, or to oppose patents of other guild members even when this has little impact on the overall industry profits. These trade-offs suggest that a well-designed opposition system needs to balance a screening and rent-preservation incentives.

4 Data and methods

Our empirical work is based on two datasets: the patents granted by the Venetian Senate and information on craft guilds active in the Venetian Republic during the Renaissance.

Our main source of data on craft guilds is the dataset "Istituzioni Corporative, Gruppi Professionali e Forme Associative del Lavoro nell'Italia Moderna e Contemporanea" (Istituzioni Corporative, henceforth) which is the outcome of a research project financed by the Italian Ministry for Education, University and Research involving multiple Italian history departments. A comprehensive description of the data is provided in Moioli (2004).

The dataset contains detailed digitized information on craft guilds in 55 Italian cities for the period 1200-1800. Key variables include the name of each guild, the time period of its activity and its geographical location. For each guild in the sample, the data provide a detailed description of the related manufacturing and trading activities. The dataset also reports a variety of indicators related to the internal organization of the guild, such as the presence of restrictions to market competition, limits for the admission of foreigners, privileges granted to offspring of the guild members, and the existence of a structured apprenticeship system.

A common restriction to competition is price fixing, but guild statutes also included additional provisions such as a minimum distance between workshops ('botteghe') or a ban to serve customers of other guild members (Moioli, 2004). Granting privileges to sons and sons-in-law of members was a typical way to restrict entry. In some cases, such as the goldsmith guild of Venice, entry was completely precluded to those who were not descendent of guild members. In other statutes, high entry fees or exams were required for those who were not sons of guild members (Moioli, 2004). Similarly, statutes sometime precluded entry to all foreigners, or to some specific ethnic group.[6]

Our main data sources on Venetian patents are the books by Roberto Berveglieri (1995, 1999) which report information on the patents granted by the Senate retrieved from the State Archives of Venice. These sources extend previous research by Giulio Mandich (1936) who translated into modern Italian and classified 109 patents granted by the Venetian Senate for the period 1474-1550. For the same period, Berveglieri (1995) describes 169 patents providing information on the technological field and the inventor. Appendix Table A1 shows the technological breakdown of these patents. Mills account for roughly half of the inventions, followed by drainage devices (11 percent) and hydraulic pumps (7 percent).

Our analysis focuses on 340 craft guilds identified in the Istituzioni Corporative dataset as active in the Venetian Republic before the year 1600. For each guild, we identify the patents granted by the Venetian Senate related to its manufacturing activities following the description provided in the Istituzioni Cooperative data. Specifically, we manually match each guild to one of the patent technology fields described in Table A1 whenever possible. For some of these guilds, we also identify a secondary technology field relevant to the activity.

The main variables used in the empirical analysis are described below.

Patents. This is the endogenous variable in the analysis. It captures the number of patents granted by the Venetian Senate from 1474 to 1550 in the primary technological field of the guild. On average, there are 1.47 patents in the main technology field in which a guild operates with a standard deviation of roughly 5 patents.

We use the following control variables to capture the main ingredients of our theoretical

[6]Unfortunately, the "Istituzioni Corporative" dataset describes the exact statutory provision only for a small subset of guilds. For most guild the information is available only as a dummy (i.e. restriction to competition? Y/N Privileges to sons? Y/N, etc...). This is the main reason why our empirical analysis focuses on binary variables.

model.

Strong internal regulation. This dummy variable is set to one if the guild has internal rules which: (i) limit competition among the members, (ii) grant privileges to sons of guild members and (iii) restrict entry rights of foreigners.

Distance to Venice. This variable captures the distance (in Kms) from the city of the guild and Venice.

Trade Guild. This dummy equals one for guilds which are only involved in trade and not in manufacturing. Roughly 46 percent of the guilds in our sample are trade guilds.

Guild Members. This information is available only for 169 guilds. On average guilds in our sample have 164 members (with median equal to 48 and standard deviation 392).

Table 1 provides summary statistics and illustrates the geographical distribution of the guilds in our sample across the main cities of the Venetian Republic. About 21 percent of the guilds in our data have strong internal regulation. Roughly 50 percent of the guilds are located in Venice. Verona, Padova and Brescia are the cities with more guilds in the mainland (Terraferma). Our main analysis focuses on the period following the patent act, 1474-1550. Restricting the study on this period allows us to avoid the 1575-76 plague which had profound impacts on the Venetian economy. Grandi (1990) and Pezzolo (2013) document the large demographic effects of the plague, with an estimated decrease in population between 15 and 26 percent.

4.1 Econometric specification

Building on the theoretical analysis of Section 3, our main econometric model focuses on the relationship between our measure of patenting, $Patents_i$, related to guild i located in city j and the indicator for the strength of internal regulation of the guild. We typically model the conditional expectation of patenting activity as

$$E(Patents_i) = \exp(\alpha \text{Strong internal regulation}_i + \beta x_i + \gamma_j)$$

where x_i is a vector of control variables and γ_j is a city-specific idiosyncratic effect. The log-link formulation is appropriate in our setting because of the non-negative and highly skewed nature of our count-based dependent variable. We estimate this model via Poisson, with robust standard errors to account for over dispersion.

5 Empirical results

Table 2 provides the first set of results. The regressions show a strong negative association between the patents granted in the technology field of a guild and the strength of its internal rules. All regressions include a control for trade guilds which indicates substantially lower patenting activity for this type of organizations. In column 2 we control for the geographic distance between the city in which a guild is located and Venice. The likelihood of patenting increases with the distance from Venice, and the coefficient on internal regulation remains stable. Column 3 shows that the relationship between guild statute strength and patenting is robust to the inclusion of city fixed effects. Exponentiation of the coefficient implies that patenting is roughly 65 percent lower when guilds adopt strong internal regulation. In column 4 we show that results are similar even when we control for the number of guild members, even if this restricts the analysis to a much smaller sample. The coefficient on the number of members is positive (but statistically insignificant), suggesting that patenting is more frequent in technology fields where guilds are larger.[7]

We perform a variety of additional empirical tests to confirm the robustness of these findings. First, we show that the estimates on the strength of internal regulation and on geographical distance are unaffected once we include additional controls for city characteristics. In column 1 of appendix table A2 we show robustness to the inclusion of controls for the size of the city measured with population in 1300, 1400 and 1500 (data from Malanima, 1998). Interestingly, population controls do not appear to explain much of the variance in patenting activity, suggesting that the number of patents is not simply driven by city size. In a model with city effects, Column 2 of table A2 shows that the negative correlation between strength of the statute and patenting is robust to including a variety of additional controls for guild characteristics such as the age of the guild (in 1500) and a dummy for the presence of an apprenticeship system.[8] The regression also includes industry effects for guilds in agriculture, construction and textile. Column 3 shows that results are similar if we use an alternative (less restrictive) mapping between guilds and patent technology sectors. More precisely, we

[7] In unreported regressions we capture guilds with a large number of members with a dummy variable equal to one if the number of members is above the top quintile (180 members). In such specifications the dummy is positive and statistically significant at the 0.1 level supporting the idea that patenting is more likely in fields where guilds have many members.

[8] De la Croix et al (2016) discuss how apprenticeship was a key determinant of economic growth in Medieval cities.

construct a new dependent variable imputing to each guild patents in the two technology sectors which are closer to the guild activity. Also in this case, results are very similar to our baseline estimates. Finally, in column 4 of table A2 we show that results are robust to expanding time period considering the patenting activity (in the main technology field) up to 1600. In a second set of unreported tests, we confirm the robustness of our main findings to estimating alternative econometric specifications such as OLS and a linear probability model for the presence of at least one patent for the guild. We also show that results are robust to dropping from the sample the guilds located in Venice.[9]

There is the concern that the effect of statutory norms on patenting is not driven by specific provisions related to entry and competition, but by other statutory rules. Specifically, the reader may worry that our '*strong internal regulation*' variable simply captures statutes which are very detailed, and that some other rule in these statutes may affect patenting more than those related to entry and competition. To address this concern, we perform a number of placebo tests constructing variables which identify statutes containing detailed regulations of guilds activities not directly related to entry and competition. For example, in column 5 of table A2 the variable '*detailed statute placebo*' equals one if the statute includes: (i) a list of manufacturing activities precluded to women, (ii) the name of the guild's patron saint and (iii) a description of the hierarchical structure of the guild. The coefficient on this variable is positive, statistically insignificant and small in magnitude. We obtain similar estimates (positive, small and statistically insignificant) with alternative placebo tests which exploit various combinations of the above variables and other statutory clauses such as the presence of an apprenticeship system, or of technical restrictions on the quality of the products. These findings support the idea that the provisions in guild statutes restricting entry and competition are strongly related to patenting propensity, but not other statutory rules.

One may also be concerned with changes in statutory clauses over time. Two things need to be noted here. First, the coding of statutory rules in the 'Istituzioni Corporative' data typically relies on documents contemporaneous to the patent act (Moioli, 2004). Thus, the

[9] We also run a regression which separately includes dummies for each of the three components of the strong guild variable (i.e., limits to price competition, privileges to offspring of the guild members and limits to foreign members). All coefficients are insignificant and we cannot reject that they are equal to each other. Including the strong guild indicator together with the three dummies also leads to insignificant coefficients for the three individual components, but a negative and statistically significant effect for strong guild with a magnitude similar to the one in our baseline regression. Moreover, dropping one of the components of the strong guild dummy leads to insignificant coefficients, but we cannot reject that they are equal to our baseline coefficient. These robustness checks suggest that it is not only one feature of the statute that drives the effect and that only guild statutes with detailed rules on all three features correlate with lower patenting.

coding is not likely to reflect the clauses present in the oldest or most modern versions of the statutes. Second, historians have emphasized how changes in guild statutes over time typically led to lower entry barriers and greater competition (Costantini, 1987). This implies that in constructing *strong internal regulation* we are more likely to classify as strong, statutes which are not strong; and thus measurement error will bias our estimates toward zero.

While the dataset provides information on whether the statute of a guild changes over time, we do not know the exact clauses which are affected by the change, which precludes us to use the longitudinal nature of the data. Nonetheless, we exploit this information to perform robustness tests. Specifically, we identify statutes which changed during the period 1474-1550. In roughly 81 percent of the sample there is no statutory change during the time period, for about 18 percent of the guilds the statute is changed once, and for the remaining 1 percent it is changed twice. In column 6 of table A2 we show that our baseline estimates are robust to dropping guilds which change their statutes during our sample period. The coefficient is roughly 15 percent larger than our baseline confirming the idea that measurement error biases our estimates toward zero.[10]

We turn next to two extensions that are of independent interest.

First, we examine whether the determinants of patenting differ comparing local and foreign inventors. We obtain information on the origin of the inventor from Berveglieri (1995) that shows that only 6.5 percent of the patents in the sample are granted to foreign inventors. Columns 1 to 4 of Table 3 show statistically significant associations between geographical distance and strength of internal rules for patenting both of local and of foreign inventors. The magnitude of these correlations is much smaller for foreign inventors. Nonetheless, our estimates show that the characteristics of the city and the guild seem to affect patenting propensity of inventors independently of their origin.[11] It is possible to reconcile the larger magnitude of the effect for local inventors with our theoretical model. In fact, under various standard assumptions on the distribution of the parameter Δ (e.g. uniform, exponential, etc.) the derivative of the probability of patenting with respect to θ has smaller magnitude for external innovators than for guild members.

[10] We confirm this result in regressions: (i) which include a control for the number of statutory changes, and (ii) consider changes over different time windows.

[11] Following Berveglieri (1995) we refer to an inventor as foreign if he is not Italian. Results are similar if we extend the definition to non-members of the Venetian Republic (about 20 percent of patents are held by foreign inventors with this definition).

Second, there is the concern that our matching between guilds and patents is done exclusively through the technology sector of each guild, ignoring the geographical scope of the patent. This is because our main data sources (Berveglieri, 1995; 1999) do not report the geographical coverage of each patent right. Indication of the geographical scope of the patent is available for the smaller sample of patents constructed by Mandich (1936). These data show that roughly 88 percent of the patents in the sample are enforceable on the entire dominion. The remaining patents are only enforceable in specific locations (e.g. only in Venice or other specific cities). We use these data to construct an alternative dependent variable which imputes the patent with limited geographical scope only to the guilds located in the relevant cities. Column 5 and 6 show that our results on the geographical distance and on the strength of internal rules are robust to exploiting this alternative measure of patenting.

5.1 Distance from Venice and patenting

The regressions reported in Tables 2 and 3 show that patenting is more pronounced in technology fields of guilds located in cities that are geographically distant from Venice. A possible interpretation of this finding is that formal protection through patent documents is more beneficial for innovators operating further away from the center of political activity. In other words, innovators who are close to Venice may have access to alternative (formal or informal) mechanisms to protect their technologies. Another interpretation of the finding is that more distant cities are inherently different in their economic activity and human capital and this explains their patenting propensity.

To examine this issue, in column 1 and 2 of Table 4 we introduce additional controls which capture additional features of the cities in our sample. First, we include the population of the city in 1500. Second, we include a measure for the number of noble families residing in each city. To construct this variable, we digitize the census of the patrician families residing in the main cities of the Veneto and nearby regions compiled in the nineteenth century by Schroeder (1830). For each family Schroeder reports the date in which the family obtains the nobility title and the city in which it resides. Exploiting this data we count the number of noble families residing in each city during our sample period. On average cities in our sample have 60 noble families with more than 100 families residing in Venice and smaller cities (such as Udine and Treviso) hosting less than 30 families.[12]

[12] To exploit this measure, we drop 6 observations from our sample because they are associated to smaller cities which were not covered in the books of Schroeder (1830).

Column 1 of Table 4 shows that there is a positive and statistically significant correlation between the number of noble families in a city and patenting by the guilds in the city. At the same time, the regression also shows that the number of noble families in a city explains much more of the variance in patents than its population. Including these controls has no effect on the estimates of the effect of geographical distance and internal strength of the guild. This finding suggests that patenting is not simply driven by the sheer size of the city, but it likely to be related to other regional characteristics. For example, the presence of noble families in a city may affect the quality of its human capital and thus spur technological activity.

In column 2 we extend our model adding a variable which captures the political strength of the noble families residing in a city. We construct this variable combining the digitized census of patrician families with multiple data sources. The first is the list of families who are members of the Great Council following the Serrata of 1297 up to the fall of the republic in 1797 provided by Raines (2004). The second is data on marriages involving a noble husband for the period 1400-1599 available from the Archivio di Stato di Venezia and digitized by Puga and Trefler (2014). Combining these data sources, we construct an indicator variable which equals one if in the city there is at least one family which belongs to the Great Council or which is linked through marriage to a family in the Great Council.[13]

Interestingly, we find a negative and statistically significant association between the presence of politically connected families and patenting of the guilds in the city, suggesting that protection through formal institutions such as patents was a substitute to alternative forms of protection available to guilds with stronger political connections. There is the concern that the results of columns 1 and 2 are driven by legal and judicial differences between Venice and other cities in the sample, because of historical traditions and because most of the noble families and members of Great Council resided in Venice (Knapton, 2013). To address this issue, in column 3 we drop from our sample all the Venetian guilds. All our findings are robust to focusing on this smaller sample of guilds located in Terraferma.

From a theoretical perspective the relationship between patenting and political or geographical distance from Venice is ambiguous. On one hand, patents might have been easier to enforce for inventors located closer to the capital. On the other hand, inventors and guild members politically connected might have been able to obtain protection from the government

[13] More than half of the cities in our sample are not connected to the Great Council according to this measure. We use an indicator variable because of the limited variance in this variable (a part from Venice, in all the other cities the number of linked families ranges between 0 and 3).

through other formal or informal channels. Our empirical evidence suggests that the second effect dominated the first one and that patents were not as widespread among guilds located in the proximity of Venice and among those with greater political connection. A variety of historical accounts can support this finding. First, the Giustizia Vecchia - which was the main magistracy enforcing guild statutes and solving disputes between guild members was located in Venice (Monticolo, 1892; Shaw, 2002). Closer interaction with guilds located in the vicinity of Venice may have allowed resolution of disputes on new products and processes without the need of formal patent documents. Second, some of the guilds located in Venice – such as those active in the Arsenal and Murano's glassblowers - were under close scrutiny of the Venetian government which often introduced legislative acts to complement guild statutes and to provide additional regulation to the sector. Therefore, technology adoption may have been regulated through alternative channels for these guilds. For example, Caniato (1996) describes various legislative acts related to the Arsenal guild members which protected local production (i.e., by burning ships not built in Venice) and which rewarded selected foreign shipbuilders (such as Antonio de Dossis from Mantova). Similarly, Davanzo Poli (1984) describes a Senate decision in 1462 with provisions supporting the tanner and shoemaker guilds of Venice.

5.2 Instrumenting guild's regulation strength

Our analysis has shown a strong negative association between the strength of the internal rules governing a guild and patenting in the technology area in which the guild operates. We documented robustness in a variety of specifications which include city effects and controls for multiple characteristics of the guilds. To interpret this correlation causally is challenging because potential unobservable variables may be correlated both with our measure of internal strength and with patenting. For example, some of the guilds in our sample may not have internal governance rules because the development of new technologies has limited their ability to set them. To assess the causal impact of statutory rules we need an instrumental variable, Z, which is correlated with the presence of internal governance and unlikely to be correlated with patenting trends.

In this section, we propose an instrument which relies on the religious origin of some of the guilds in our dataset. In fact, a number of the guilds in our sample can be linked to religious confraternities. These confraternities originated from the success of the flagellant movement of the 13th century, which preceded the diffusion of artisan guilds in the Venetian Republic.[14]

A confraternity (also called 'scola' or 'fratalea') was an association of lay people driven by Christian devotion and work of charity (Gasparini, 1987). Confraternity members were required to follow rules and bylaws in exchange of help in time of hardship and the security of a good funeral (Moniticolo, 1905). While people from all social classes could join a confraternity, most of the members were artisans. Typically, each confraternity attracted artisans from the same profession, which often led to the formation of a guild linked to the confraternity (Gasparini, 1987). For example, in Venice the guild of ironmongers was linked with the confraternity of San Lorenzo, fishermen with that of San Nicolo' and goldsmiths with that of San Mattio (Mackenney, 1994).[15]

The 'Istituzioni Corporative' data show that roughly 30 percent of the guilds in our sample are linked to a religious confraternity. These guilds are scattered across the various cities and cover a variety of activities. For example, the barbers' guild in Verona derives from a confraternity, but none of the barber guilds of the other cities in the sample appear to have religious linkages. Similarly, the blacksmiths' guild of Udine is related to a confraternity whereas those of Padova, Venezia and Vicenza are not. There is some geographical variation in the religious origin of guilds. More than half of the guilds in Venice are linked to a confraternity whereas in the other cities the proportion is typically below 20 percent. Once we control for city effects, we do not find any significant correlation between the religious origin of the guild and other observable guild characteristics such as its age or the presence of an apprenticeship system. Table A3 in the appendix documents such low correlations with a series of regressions with religious links as dependent variable. This suggests that the religious origin of the guild is likely to be driven by idiosyncratic reasons, related to the local success of the flagellant movement in the 13th century, and thus unlikely to be correlated with shocks affecting patenting trends two centuries in the future. At the same time, because members of religious institutions were governed by a precise set of rule, which was generally recorded in a book called 'Mariegola', it is reasonable to expect guilds with religious origin to also have stronger internal regulations. Mackenney (1994) explains how guild statutes often were inspired by rules of the related confraternities.

[14] Our data show that about 80 percent of the guilds originated in the late 14th and 15th century.

[15] These links generated obligations from both sides. For example, guilds were required to make financial contribution to the confraternity, but were also allow to use the confraternity venues as meeting places.

In Table 5 we exploit the linkages between guilds and confraternities as instrumental variable. Column 1 reports the first stage regression, which indicates a strong positive correlation between the religious origin of the guild and the strength of its internal rules. Columns 2 and 3 contrast the OLS estimates and the 2SLS estimates of similar linear regression models. Both specifications confirm the strong negative relationship between patenting and guild's internal rules. The estimates of the IV regression are larger in magnitude but not statistically different from those in the OLS model. The larger magnitude of the coefficient is consistent with measurement error in statutory strength biasing the estimates toward zero, as discussed above.[16]

Following Galasso et al (2013) we also instrumented the '*strong internal regulation*' dummy with the predicted probability of a strong statute obtained from a probit model. When the endogenous regressor is a dummy, this estimator is asymptotically efficient in the class of estimators where instruments are a function of the religious origin of the guild and other covariates (Wooldrige, 2002). The 2SLS estimate with this alternative model is essentially identical in magnitude and significance level as the one presented in column 3 of Table 5.

6 Conclusions

The Venetian act of 1474 which regulated the patent granting process provides a unique opportunity to study the diffusion of a new economic institution. Motivated by a theoretical model which illustrates the role of guild statutes on the propensity to rely on patents, our paper documents a strong negative association between the number of patents granted in the technology sector of a guild and the presence of statutory provisions limiting entry and price competition. We also show that patenting is more frequent in cities which are geographically distant from Venice and in cities with less political connection, which we measure exploiting a unique database with location of noble families and their marriages with members of the Grand Council.

Overall, our findings indicate that the diffusion of new economic institutions may be strongly affected by features of the local economic and political environment. This has potential implications for the design of innovation policies, because it suggests that the success of a policy may vary substantially across locations and industries, even within a region. Our estimates

[16] Following Angrist and Pischke (2009), we exploit the first stage estimates to compute the proportion of the treated who are compliers which is 0.22. This indicates that our estimates are not specific to a small compliant subpopulation.

are also in line with the more recent innovation literature which has documented substantial variation in the rate of patenting across industries and in the perceived effectiveness of patents across firms (Levin et al., 1987; Cohen, Nelson, and Walsh, 2000). Our data show that even in the very first patent system the private economic value provided by intellectual property rights appeared highly heterogeneous. Finally, our analysis underscores the importance of considering the potential substitution between new institutions and existing alternatives.

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Appendix: Extensions of the theoretical model

Generalized impact of guild statutes

In this Appendix we extend our baseline model generalizing the impact that guild statutes have on rent sharing, $\alpha(\theta)$, and entry, $\beta(\theta)$. More specifically we consider functions $\alpha'(\theta) > 0$ and $\beta'(\theta) < 0$ with $\alpha(0) = \beta(1) = 0$ and $\alpha(1) = \beta(0) = 1$.

We start analysing the decision to patent and oppose a patent in the case in which a guild member is the innovator. Patenting is more profitable than not patenting only if

$$\pi + \Delta - \frac{\alpha(\theta)\pi}{2} + \frac{\alpha(\theta)(\pi + \Delta)}{2} - c > \frac{2\alpha(\theta)(\pi + \Delta)}{2}$$

or

$$\Delta > \widetilde{\Delta} = \frac{2\left(c - \pi(1 - \alpha(\theta))\right)}{2 - \alpha(\theta)}.$$

Notice that $\widetilde{\Delta} > 0$ only if $\alpha(\theta)$ is large enough. Moreover $\frac{d\widetilde{\Delta}}{d\alpha} > 0$ which combined with our assumption that $\alpha'(\theta)$ implies that, as the strength of the internal statute increases, guild members patent only their more valuable innovations and the propensity to patent decreases in θ. Consider now the incentives of one guild member to oppose the patent by the other guild member. Opposition is profitable if

$$\frac{2\alpha(\theta)(\pi + \Delta)}{2} - \kappa > \frac{\alpha(\theta)\pi}{2} + \frac{\alpha(\theta)(\pi + \Delta)}{2}$$

which is satisfied if

$$\Delta > \widehat{\Delta} = \frac{2}{\alpha(\theta)}\kappa.$$

Notice that $\frac{d\widehat{\Delta}}{d\alpha} < 0$ which combined with $\alpha'(\theta)$ implies that, as the strength of the internal statute increases, guild members block patents of other guild members more often. This implies that our results on patenting by guild members are robust to assuming a more general relationship between rent-sharing and θ.

Conside now the case of an external innovator. Patenting is more profitable than entry without patent if

$$\pi + \Delta - \alpha(\theta)\pi + \frac{\alpha(\theta)(\pi + \Delta)}{3} - c > \beta(\theta)\frac{2\alpha(\theta)(\pi + \Delta)}{3}$$

that occurs if

$$\Delta > \widetilde{\Delta}_E = \frac{3c - 3\pi + 2\pi\alpha(\theta) + 2\pi\alpha(\theta)\beta(\theta)}{\alpha(\theta) - 2\alpha(\theta)\beta(\theta) + 3}$$

Notice that because $\alpha(0) = \beta(1) = 0$ and $\alpha(1) = \beta(0) = 1$ $\widetilde{\Delta}_E(0) = (3c - 3\pi)/3$ and $\widetilde{\Delta}_E(1) = (3c - \pi)/4$ which are both negative because $\pi > 3c$. Moreover, we have that

$$\begin{aligned}\frac{d\widetilde{\Delta_E}}{d\theta} &= 3\frac{3\pi - c + 2\beta(\theta)c}{(\alpha(\theta) - 2\alpha(\theta)\beta(\theta) + 3)^2}\alpha'(\theta) \\ &\quad +6\alpha(\theta)\frac{c + \pi\alpha(\theta)}{(\alpha(\theta) - 2\alpha(\theta)\beta(\theta) + 3)^2}\beta'(\theta)\end{aligned}$$

and a condition to have $\frac{d\widetilde{\Delta_E}}{d\theta} \geq 0$ is

$$-\frac{\alpha'()}{\beta'()} \geq \frac{2(\alpha(\theta)(c + \pi\alpha(\theta)))}{3\pi - c + 2\beta(\theta)c}$$

notice that the right hand side is bounded by $(2c+2\pi)/(3\pi - c)$ which in turn is bounded by 1 because $\pi > 3c$. This implies that a sufficient condition to have $\frac{d\widetilde{\Delta_E}}{d\theta} \geq 0$ is that $|\alpha'()| \geq |\beta'()|$, i.e. changes in statute have greater impact on rent sharing than on entry.

Finally, for guild members opposing the patent is more profitable than accommodating if

$$2\alpha(\theta)\frac{(\pi + \Delta)}{2} - \kappa > \frac{\pi}{2}\alpha(\theta) + \frac{\alpha(\theta)(\pi + \Delta)}{3}$$

or

$$\Delta > \widehat{\Delta_E}(\theta) = \frac{3}{2\alpha(\theta)}\left(\kappa - \frac{\alpha(\theta)\pi}{6}\right)$$

which is decreasing in α which combined with our assumption that $\alpha'(\theta) > 0$ implies that the likelihood of opposition increases in θ.

Generalized licensing negotiations

In the baseline setting, we assume that the innovating firm has the full bargaining power during the licensing negotiations and that it appropriates the whole surplus of the innovation (while the other guild members obtain the status-quo profits $\theta\pi/2$). In this Appendix, we generalize the analysis assuming that the surplus is shared according to parameter $\gamma \in [0, 1]$. More specifically, during the period of validity of the patent, the innovating firm obtains its status–quo profits plus a share γ of the innovation surplus, $\pi + \Delta - \theta\pi$; the remaining $(1-\gamma)$ share is appropriated by the other guild member(s). Parameter γ represents the bargaining power of the inventor during the licensing negotiations. Notice that when $\gamma = 1$ we are back to the baseline setting.

Below, we show that the comparative static results that we show for the baseline setting still hold true in this more general framework provided that γ is large enough. Consider first the case of innovation by a guild member. By patenting, the innovator obtains

$$\frac{\theta\pi}{2} + \gamma(\pi + \Delta - \theta\pi) + \frac{\theta(\pi + \Delta)}{2} - c.$$

In the first period, the firm obtains the status-quo profits, $\theta\pi/2$, plus the share γ of the innovation surplus $(\pi + \Delta - \theta\pi)$. Hence, patenting is more profitable than non-patenting only if

$$\frac{\theta\pi}{2} + \gamma(\pi + \Delta - \theta\pi) + \frac{\theta(\pi + \Delta)}{2} - c > \frac{2\theta(\pi + \Delta)}{2}$$

or

$$\Delta > \tilde{\Delta} = \frac{2\left(c - \gamma\pi(1-\theta)\right)}{(2\gamma - \theta)}.$$

Notice that $\tilde{\Delta} > 0$ provided that θ and γ are large enough; moreover, $d\tilde{\Delta}/d\theta > 0$, which implies that the propensity to patent reduces with the strength of the statutes.

Consider now the incentives to oppose the patent by a guild member. Opposition is profitable if

$$\frac{2\theta(\pi + \Delta)}{2} - \kappa > \frac{\theta\pi}{2} + (1-\gamma)(\pi + \Delta - \theta\pi) + \frac{\theta(\pi + \Delta)}{2}.$$

When choosing not to oppose the patent, during the first period, the guild member obtains $\theta\pi/2 + (1-\gamma)(\pi + \Delta - \theta\pi)$, the status-quo profits plus a share $(1-\gamma)$ of the innovation surplus. Hence, opposition is optimal if

$$\Delta > \hat{\Delta} = \frac{2\kappa + 2(1-\gamma)(1-\theta)\pi}{(\theta - 2(1-\gamma))}.$$

A simple inspection of $\hat{\Delta}$ reveals that $d\hat{\Delta}/d\theta < 0$: the larger θ the more likely that guild members oppose patents by other guild members. Let us focus now on the case of innovation by a non-guild member. Patenting is profitable if

$$\gamma(\pi + \Delta - \theta\pi) + \frac{\theta(\pi + \Delta)}{3} - c > (1-\theta)\frac{2\theta(\pi + \Delta)}{3},$$

or

$$\Delta > \tilde{\Delta}_E = \frac{3c - \pi\left(3\gamma(1-\theta) - \theta + 2\theta^2\right)}{(3\gamma - \theta + 2\theta^2)}.$$

Notice that $\tilde{\Delta}_E < 0$ if $\gamma > \frac{3c+\pi\theta(1-2\theta)}{3\pi(1-\theta)}$; hence, the external innovator always prefers to patent provided that γ is large enough.

Finally, for guild members opposition to patents granted to external innovators is profitable when

$$\frac{2\theta(\pi + \Delta)}{2} - \kappa > \frac{\theta\pi}{2} + \frac{(1-\gamma)}{2}(\pi + \Delta - \theta\pi) + \frac{\theta(\pi + \Delta)}{3}$$

Notice, when accommodating the patent, during the first period, each guild member obtains its status-quo profits plus half of the share of the innovation surplus going to non-inventors, i.e. half of $(1-\gamma)(\pi + \Delta - \theta\pi)$. Therefore, a guild member chooses to oppose a patent by an external if

$$\Delta > \hat{\Delta}_E = \frac{3c - \pi\left(3\gamma(1-\theta) - \theta + 2\theta^2\right)}{(3\gamma - \theta + 2\theta^2)}.$$

One can easily check that $d\hat{\Delta}_E/d\theta < 0$ which implies that the larger θ the more likely that guild members oppose patents by external innovators

Figure 1. Venetian State Boundary

Map 3.1. Venice's *Terraferma*.

(Source: Knapton, 2013)

Table 1. Summary statistics

	Obs.	Mean	Std. Dev.	Min	Max
Patents	340	1.47	5.10	0	42
Strong internal regulation	340	0.21	0.40	0	1
Distance to Venice (km)	340	56.35	72.48	0	475
Trade guild	340	0.46	0.49	0	1
Guild members	169	164.06	392.27	2	3390
Main Cities					
Venice	161				
Verona	54				
Padova	39				
Brescia	31				
Treviso	16				
Udine	13				
Others	26				

NOTES: Unit of observation is a guild. Patents is the total number of patents granted from 1474 to 1550 in the technology sector of the guild. Strong internal regulation =1 if guild has internal rules which restrict competition, grant privileges to sons of members, and restrict rights of foreign members. Trade guild is equal to one if the guild is not involved in manufacturing. Guild members = number of registered members as reported in the "Istituzioni Cooperative" data.

Table 2 . Guild internal regulation and patenting

	(1)	(2)	(3)	(4)
Dependent Variable	Patents	Patents	Patents	Patents
Strong internal regulation	-0.750***	-0.981***	-1.133***	-1.717***
	(0.151)	(0.101)	(0.439)	(0.610)
log(Distance)		0.225***		
		(0.026)		
log (Guild members)				0.256
				(0.191)
Trade guild	-4.535***	-4.357***	-4.355***	-5.268***
	(0.710)	(0.708)	(0.848)	(1.042)
City Effects	No	No	Yes	Yes
Observations	340	340	340	169

NOTES: Poisson estimation with robust standard errors. * significant at 10 percent, ** significant at 5 percent and *** significant at 1 percent. Strong internal regulation =1 if guild has internal rules which restrict competition, grant privileges to sons of members, and restrict rights of foreign members. Trade guild is equal to one if the guild is not involved in manufacturing. Distance= distance from Venice in Km. Guild members = number of registered members as reported in the "Istituzioni Cooperative" data.

Table 3. Inventors' origin and alternative patent data

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable	Patents Local	Patents Local	Patents Foreigners	Patents Foreigners	Patents Mandich	Patents Mandich
Strong internal regulation	-0.946***	-1.098**	-0.029**	-0.034*	-1.363***	-1.266**
	(0.095)	(0.429)	(0.012)	(0.020)	(0.313)	(0.616)
log(Distance)	0.219***		0.006***		0.170***	
	(0.025)		(0.001)		(0.036)	
City Effects	No	Yes	No	Yes	No	Yes
Observations	340	340	340	340	340	340

NOTES: Poisson estimation with robust standard errors. * significant at 10 percent, ** significant at 5 percent and *** significant at 1 percent. All regressions include a dummy for Trade guilds. Strong internal regulation =1 if guild has internal rules which restrict competition, grant privileges to sons of members, and restrict rights of foreign members. Distance= distance from Venice in Km. Local patents = patents granted to Italian inventors. Foreign patents = patents granted to non-Italian inventors. Columns 1-4 exploit patent data from Berveglieri (1995, 1999) columns 5-6 exploit patent data from Mandich (1936).

Table 4. Noble families and patenting

	(1)	(2)	(3)
Dependent Variable	Patents	Patents	Patents
Strong internal regulation	-1.143***	-1.203***	-1.283***
	(0.120)	(0.125)	(0.130)
log(Distance)	0.326***	0.340***	0.264**
	(0.029)	(0.022)	(0.118)
log(Noble Families)	0.103***	0.419***	0.456***
	(0.027)	(0.093)	(0.098)
log($Population_{1500}$)	0.167	0.027	0.080
	(0.116)	(0.072)	(0.107)
Politically Connected Families		-1.333***	-1.500***
		(0.351)	(0.398)
City Effects	No	No	No
Drop Venice	No	No	Yes
Observations	334	334	173

NOTES: Poisson estimation with robust standard errors. * significant at 10 percent, ** significant at 5 percent and *** significant at 1 percent. All regressions include a dummy for Trade guilds. Strong internal regulation =1 if guild has internal rules which restrict competition, grant privileges to sons of members, and restrict rights of foreign members. Distance= distance from Venice in Km. Noble families = number of noble families in the city as registered by Schroeder (1830). Population= inhabitants in 1500 as estimated by Malanima (1998). Politically connected families=1 if there is at least one family in the city which belongs to the Great Council or is linked through marriages to a family in the Great Council.

Table 5. Religious confraternities and guild internal strength

	(1)	(2)	(3)
Dependent Variable	Strong guild	Patents	Patents
Estimation	OLS	OLS	2SLS
Religious confraternity	0.150***		
	(0.045)		
Strong internal regulation (instrumented)		-1.958**	-4.183*
		(0.861)	(2.720)
City Effects	Yes	Yes	Yes
Observations	340	340	340
First stage F-test			7.85

NOTES: OLS estimation with robust standard errors. * significant at 10 percent, ** significant at 5 percent and *** significant at 1 percent. All regressions include a dummy for Trade guilds. Strong internal regulation =1 if guild has internal rules which restrict competition, grant privileges to sons of members, and restrict rights of foreign members. Religious confraternity =1 if guild is linked to a religious institution.

Table A1. Patents by technology sector 1474-1550

Mills	
cereals	42
metals	6
textiles	9
wood saws	6
multiple usage	22
Fabrics	8
Paint	4
Bread and food	1
Pottery and Porcelain Vases	1
Agricultural machines	4
Drainage, mud removal	20
Hydraulic pumps	11
Guns	7
Arsenal	3
Mining	3
Perpetual Motion	3
Miscellaneous	19
Total	169

Source: Berveglieri (1995)

Table A2. Robustness

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable	Patents	Patents	Patents Sector 1 & 2	Patents up to 1600	Patents	Patents
Strong internal regulation	-1.012***	-0.948**	-0.940***	-0.728**		-1.307***
	(0.101)	(0.431)	(0.307)	(0.318)		(0.486)
Detailed statute placebo					0.282	
					(0.394)	
log(Distance)	0.274***					
	(0.074)					
log ($Population_{1500}$)	0.038					
	(0.471)					
log ($Population_{1400}$)	0.188					
	(0.784)					
log($Population_{1300}$)	-0.072					
	(0.383)					
Apprenticeship		0.120				
		(0.307)				
log(Age)		0.047				
		(0.164)				
City effects	No	Yes	Yes	Yes	Yes	Yes
Industry effects	No	Yes	No	No	No	No
Drop guilds with change in statute	No	No	No	No	No	Yes
Observations	340	340	340	340	340	275

NOTES: Poisson estimation with robust standard errors. * significant at 10 percent, ** significant at 5 percent and *** significant at 1 percent. Regressions include dummy for Trade guilds. Strong internal regulation =1 if guild has internal rules which restrict competition, grant privileges to sons of members, and restrict rights of foreign members. Trade guild is equal to one if the guild is not involved in manufacturing. Distance= distance from Venice in Km. Population data are from Malanima (1998). Apprenticeship=1 if the "Istituzioni Corporative" database documents an apprenticeship requirement. Age= age of the guild in 1500. Industry effects are dummies for guilds in agricolture, textile and construction. Detailed statute placebo=1 if the statute includes: (i) a list of manufacturing activities precluded to women, (ii) the name of the guild's patron saint and (iii) a description of the hierarchical structure of the guild. Column (6) drops guilds with changes in statute in the period 1474-1550.

Table A3. Exogeneity of religious linkages

	(1)	(2)	(3)	(4)
Dependent Variable	Religious confraternity	Religious confraternity	Religious confraternity	Religious confraternity
Age	0.001			0.001
	(0.001)			(0.001)
Age^2	-0.001			-0.001
	(0.001)			(0.001)
Apprentship		0.046		0.042
		(0.042)		(0.042)
Number of statutory changes 1474-1550			0.022	0.039
			(0.044)	(0.044)

NOTES: OLS regression with robust standard errors. * significant at 10 percent, ** significant at 5 percent and *** significant at 1 percent. The dependent variable is filtered with city effects and a dummy for trade guilds. Apprenticeship=1 if the "Istituzioni Corporative" database documents an apprenticeship requirement. Age= age of the guild in 1500.